Do Now · retrieval

Arrays, records and tuples

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Why must exponents match before adding binary mantissas?

Q2

What does passing an int by value mean?

Q3

What is values[1] for int[] values={4,6,9}?

Q4

a={1,2,3}; a[0]=a[1]+a[2]. Find a.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Zero-based indexing — scores[0] is 4 and scores[2] is 2.

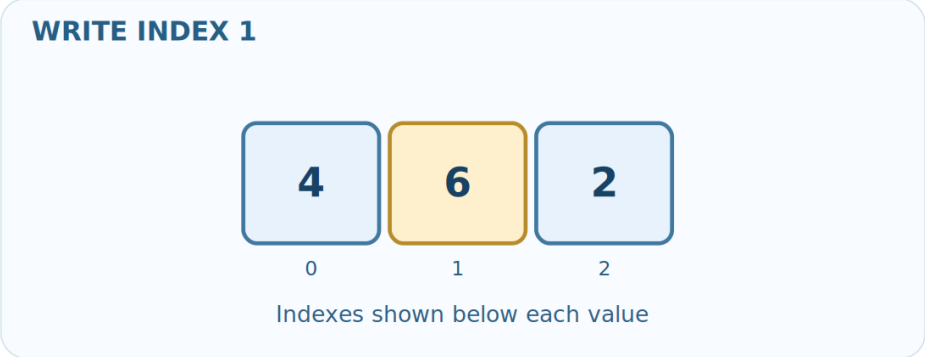
Core idea 2

Read before writing — The right-hand expression is 4 + 2 = 6.

Core idea 3

Replace the selected value — Assign 6 at index 1; the array now contains 4, 6, 2.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

array — A fixed-length indexed collection of values of one element type in C#.

index — Zero-based position of an array element.

record — A structure that groups related named fields or properties.

tuple — A compact ordered grouping of multiple values.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

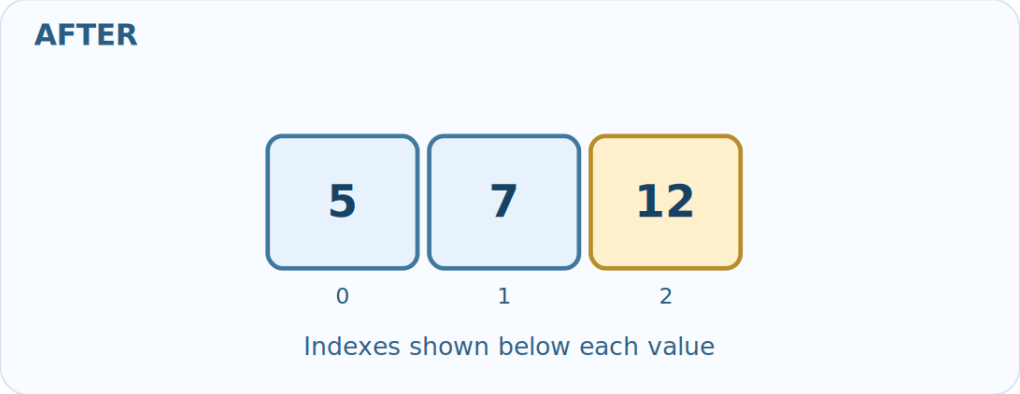
Problem

Trace marks = {5,7,3}; marks[2] = marks[0] + marks[1].

Follow the worked method

1. Predict — marks[0] + marks[1] means 5 + 7.
2. Confirm and explain — 12 replaces the value at index 2: [5,7,12].

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict

a[0] is 3, so 3 × 2 = 6.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Give the final array after int[] a={8,4,2}; a[1]=a[0]-a[2];

Question 2

When would you choose a record rather than a simple array for a pupil's name and mark?

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain a suitable use of a named tuple in C# and one limitation.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

An array with three values has a last index of 3.

Use key vocabulary from memory

1. Explain array in your own words.

2. Explain index in your own words.

3. Explain record in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.